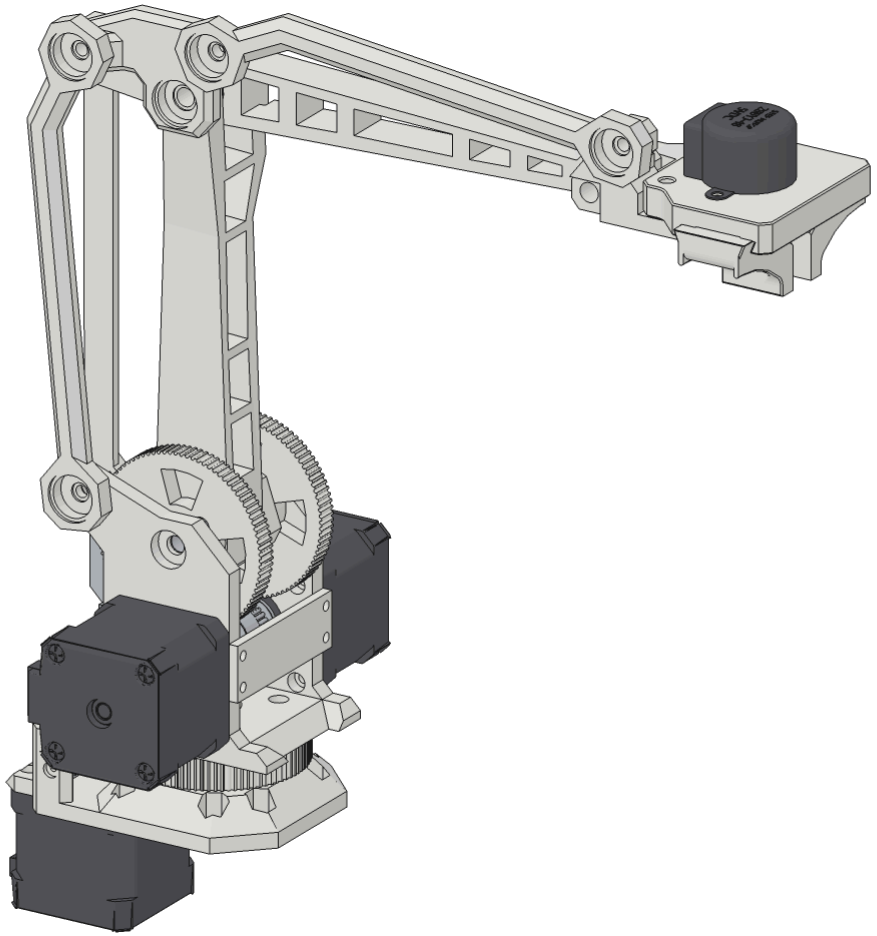


Mechy Demo bam-v2

4-DOF BELT-DRIVEN CATCHING ARM

REACH
0.334 m computed

PRINTED PARTS
21 pcs / 206 g estimate

PURCHASED PARTS
28 lines / $131 estimate

ASSEMBLY
7 steps

NOTES

[1] Belt-driven 4-DOF arm (base 5.625:1, shoulder 5.625:1, elbow 5.625:1, wrist 4.5:1); every motor sits low, joints turn on 608-2RS / 688-2RS bearings, GT2 closed belts.

[2] Built from design/bam-v2/params.yaml by mechy cad build bam-v2; numbers on these sheets come from the project files named on each sheet.

[3] Estimates are labelled. Read the last sheet (checks, not verified, open decisions) before buying.

MECHY					
					SHEET TITLE: BUILD GUIDE BAM-V2
	PROJECT: Mechy Demo	VARIANT: BAM-V2	SCALE: NTS	DATE: 2026-10-30	SHEET: 1 / 33

2 Contents and files

CONTENTS

SHEET	TITLE
1	Build guide BAM-V2
2	Contents and files
3	Bill of materials
4	Print list
5	Part arm_socket
6	Part arm_rotategear
7	Part arm_main_body
8	Part arm_gear_body
9	Part arm_endstop_bracket
10	Part arm_stabilizer
11	Part arm_lever
12	Part arm_lower_shank_140
13	Part arm_upper_shank_140
14	Part arm_pleuel_140
15	Part arm_pleuel_bend_lower_140
16	Part arm_pleuel_bend_upper_140
17	Part arm_triplate
18	Part arm_manipulator
19	Part tool_shank_calibrator_140
20	Part grip_base
21	Part grip_finger
22	Part grip_shaft
23	Part grip_screw_secure
24	Step 1 Prepare the printed parts
25	Step 2 Socket and base yaw drive
26	Step 3 Main body, motors and belts
27	Step 4 Linkage: shanks, lever, pleuel and levelling links
28	Step 5 Toolhead
29	Step 6 Electronics
30	Step 7 Home and calibrate
31	Belts, tensioning and homing
32	Wiring and flashing
33	Checks, not verified, open decisions

WHAT | WHERE

WHAT	WHERE	REBUILD
Design parameters	design/bam-v2/params.yaml	edit, then the commands below
Assembly CAD	design/bam-v2/arm.step, arm.glb, arm.stl	mechy cad build bam-v2
Printable parts	design/bam-v2/stl/*.stl (19 files)	mechy cad build bam-v2
Print list	design/bam-v2/print_list.csv	mechy cad build bam-v2
Bill of materials	design/bam-v2/bom.csv	mechy cad build bam-v2
Mass properties, belts, homing	design/bam-v2/mass_props.json	mechy cad build bam-v2
This guide	docs/bam-v2/build_guide.pdf (.typ, .data.json, img/)	mechy doc build bam-v2
Design report	docs/bam-v2/report.pdf	mechy doc build bam-v2

HOW TO READ THIS GUIDE

[1] Sheets are A4 landscape. Part sheets are true to scale when printed at 100% (scale in the title block); dimensions are in mm.

[2] Third-angle projection: TOP above FRONT, RIGHT beside it. The iso view is not to scale.

[3] Assembly sheets: blue parts are added in that step (moved out a little for clarity).

[4] Every number is computed, from a datasheet, or labelled an estimate; the source file is named on each sheet.

MECHY

SHEET TITLE
CONTENTS AND FILES

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	NTS	2026-10-30	2 / 33

3 Bill of materials

DESIGN/BAM-V2/BOM.CSV · TOTAL = SUM OF QTY X UNIT PRICE

#	ITEM	QTY	SPEC	SOURCE	USD EA	USD
1	Arduino Uno R3	1	ATmega328P, USB-B (user's)	user's stock / BAM bom	12.00	12.00
2	CNC Shield V3	1	Protoneer-style, 4 driver sockets X/Y/Z/A (user's)	user's stock / BAM bom	5.00	5.00
3	A4988 stepper driver	3	with heatsink; Vref for ~0.8 A (user's)	user's stock / BAM bom	2.00	6.00
4	A4988 stepper driver (spare)	1	A socket	user's stock / BAM bom	2.00	2.00
5	NEMA 17 stepper 42SHDC3026-15B	3	bipolar 4-wire, 0.9 A/phase, ~0.30 N·m holding (ASSUMED)	user's stock / BAM bom	12.00	36.00
6	NEMA 17 stepper 42SHDC3026-15B (spare)	1	4th motor owned	user's stock / BAM bom	12.00	12.00
7	Power supply 24 V 6 A	1	user's; A4988 VMOT <= 35 V so 24 V is fine	user's stock / BAM bom	18.00	18.00
8	Electrolytic capacitor 100 uF 50 V	1	across the shield's motor power input (user's)	user's stock / BAM bom	0.30	0.30
9	GT2 pulley 20T	3	6 mm belt, 5 mm bore	user's stock / BAM bom	1.50	4.50
10	GT2 closed belt 200 mm	3	6 mm wide closed loop	user's stock / BAM bom	2.00	6.00
11	51105 thrust bearing	1	25x42x11	user's stock / BAM bom	3.00	3.00
12	F624ZZ flanged bearing	6	4x13x5	user's stock / BAM bom	0.40	2.40
13	F686ZZ flanged bearing	12	6x13x5	user's stock / BAM bom	0.40	4.80
14	Endstop microswitch	3	NO to GND	user's stock / BAM bom	0.50	1.50
15	Servo MG90S (gripper toolhead)	1	9 g metal gear; only with toolhead: gripper	user's stock / BAM bom	3.00	3.00
16	M2x10 screw + nut	6		user's stock / BAM bom	0.10	0.60
17	M3x6 screw	36		user's stock / BAM bom	0.10	3.60
18	M3 nut	8		user's stock / BAM bom	0.05	0.40
19	M4x10 screw	6		user's stock / BAM bom	0.10	0.60
20	M4x16 screw	8		user's stock / BAM bom	0.10	0.80
21	M4x20 screw	6		user's stock / BAM bom	0.10	0.60
22	M4 nut	14		user's stock / BAM bom	0.05	0.70
23	M6x20 screw	1		user's stock / BAM bom	0.20	0.20
24	M6x40 screw	2		user's stock / BAM bom	0.20	0.40
25	M6x50 screw	1		user's stock / BAM bom	0.20	0.20
26	M6 nut	4		user's stock / BAM bom	0.05	0.20
27	Stepper cable	3	4-wire, 1 m	user's stock / BAM bom	1.50	4.50
28	Silicone wire 18 AWG	2	red/black, 1 m	user's stock / BAM bom	1.00	2.00
	TOTAL: 28 LINES	**138**	**pieces incl. spares; prices are estimates**			**131.30**

4 Print list

DESIGN/BAM-V2/PRINT_LIST.CSV

PART	QTY	MATERIAL	ORIENTATION	G EA	G TOTAL	SHEET
arm_socket	1	PLA	open side up	24.8	24.8	5
arm_rotategear	1	PLA	gear face down	11.4	11.4	6
arm_main_body	1	PLA	flat on the bed	29.0	29.0	7
arm_gear_body	2	PLA	flat on the bed	8.7	17.3	8
arm_endstop_bracket	1	PLA	flat on the bed	3.7	3.7	9
arm_stabilizer	1	PLA	flat on the bed	2.6	2.6	10
arm_lever	1	PLA	flat on the bed	5.2	5.2	11
arm_lower_shank_140	1	PLA	flat, 140 mm	27.3	27.3	12
arm_upper_shank_140	1	PLA	flat, 140 mm	19.6	19.6	13
arm_pleuel_140	1	PLA	flat	7.1	7.1	14
arm_pleuel_bend_lower_140	1	PLA	flat	7.2	7.2	15
arm_pleuel_bend_upper_140	1	PLA	flat; BAM-made part (CC BY 4.0): the vendor 140 mm kit has no upper levelling link	7.2	7.2	16
arm_triplate	1	PLA	flat	5.2	5.2	17
arm_manipulator	1	PLA	flat	8.5	8.5	18
tool_shank_calibrator_140	1	PLA	flat; homing aid	7.3	7.3	19
grip_base	1	PLA	for the MG90S / 9 g servo	12.1	12.1	20
grip_finger	2	PLA	flat	4.5	9.0	21
grip_shaft	1	PLA	flat	1.0	1.0	22
grip_screw_secure	1	PLA	flat	0.3	0.3	23
TOTAL	**21**		**19 STL files**		**205.8**	

PRINTER SETTINGS

SETTING	VALUE
material	PLA
infill	25% (grid or gyroid)
walls / top / bottom	0.8 mm per face (2 perimeters of a 0.4 mm nozzle)
layer height	0.2 mm (suggested; the CAD does not depend on it)
supports	none expected in the listed orientation (check the bearing bores)
nozzle / bed	PETG: about 235 / 80 C (typical; use your filament's values)

[1] Grams are computed from each STL's volume x PLA density x a shell/infill factor (estimate until weighed).

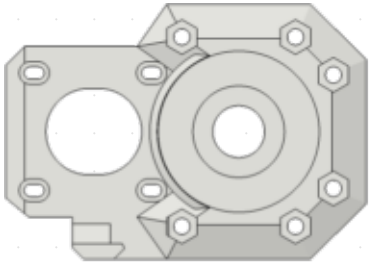
[2] Orientation is chosen for strength along the load path and bearing-seat roundness; keep it.

MECHY

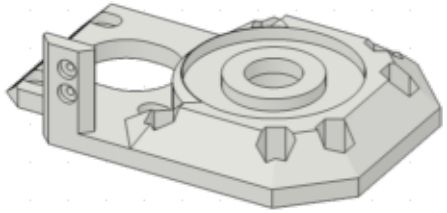
SHEET TITLE: PRINT LIST

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	NTS	2026-10-30	4 / 33

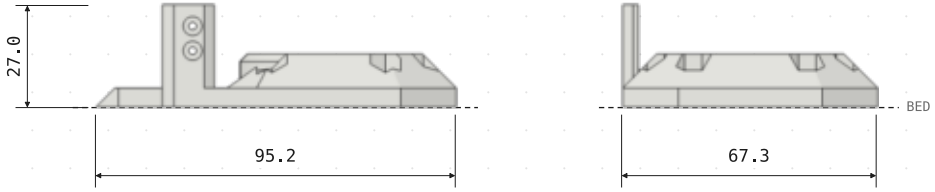
5 Part arm_socket



TOP



ISO (NOT TO SCALE)



FRONT

RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_socket.stl
qty	1
material	PLA
infill	25%
orientation	open side up
size x y z	95.2 x 67.3 x 27.0 mm
volume	41.1 cm3
mass ea	24.8 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, open side up. The dashed BED line is the print bed.

[2] Overall size 95.2 x 67.3 x 27.0 mm in the print orientation (measured from the STL).

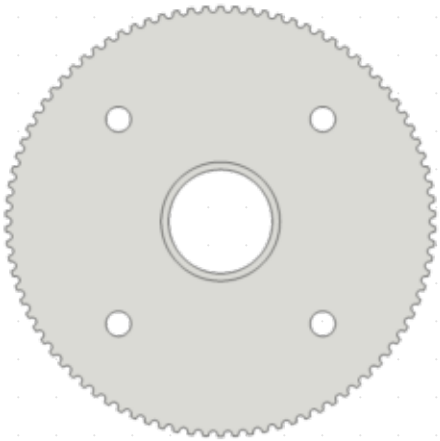
[3] Mass 24.8 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

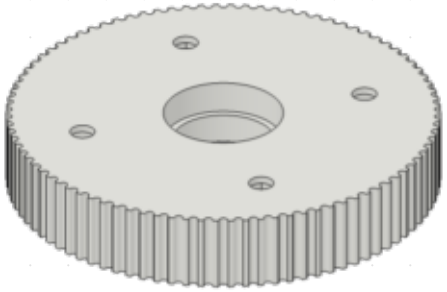
SHEET TITLE
PART ARM_SOCKET

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:2	2026-10-30	5 / 33

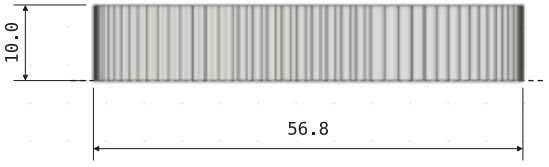
6 Part arm_rotategear



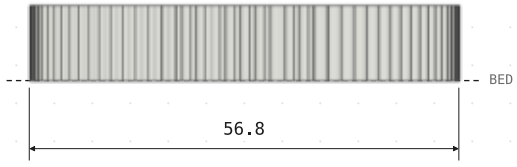
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_rotategear.stl
qty	1
material	PLA
infill	25%
orientation	gear face down
size x y z	56.8 x 56.8 x 10.0 mm
volume	16.0 cm3
mass ea	11.4 g (estimate)
scale	1:1 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, gear face down. The dashed BED line is the print bed.

[2] Overall size 56.8 x 56.8 x 10.0 mm in the print orientation (measured from the STL).

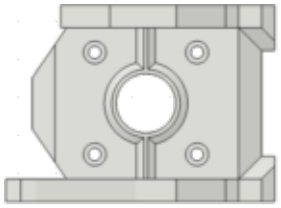
[3] Mass 11.4 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

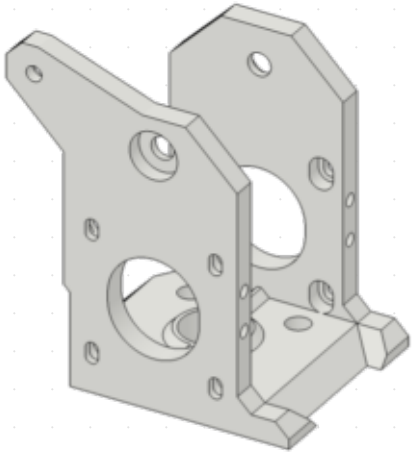
SHEET TITLE
PART ARM_ROTATEGEAR

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:1	2026-10-30	6 / 33

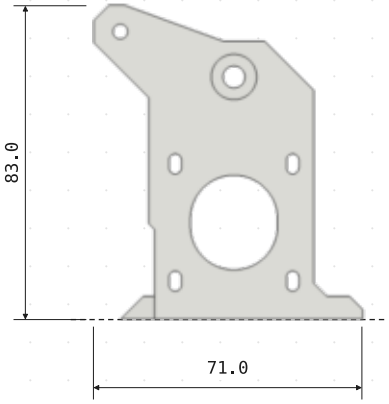
7 Part arm_main_body



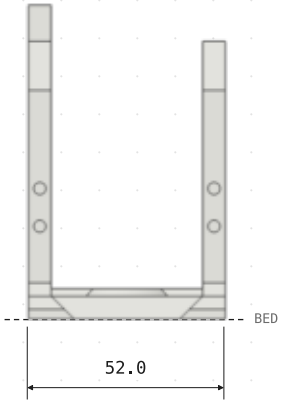
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_main_body.stl
qty	1
material	PLA
infill	25%
orientation	flat on the bed
size x y z	71.0 x 52.0 x 83.0 mm
volume	42.3 cm3
mass ea	29.0 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat on the bed. The dashed BED line is the print bed.

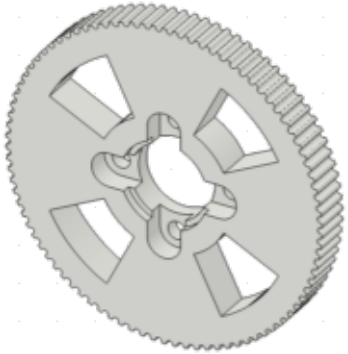
[2] Overall size 71.0 x 52.0 x 83.0 mm in the print orientation (measured from the STL).

[3] Mass 29.0 g each is computed from volume x density x shell/infill factor (estimate until weighed).

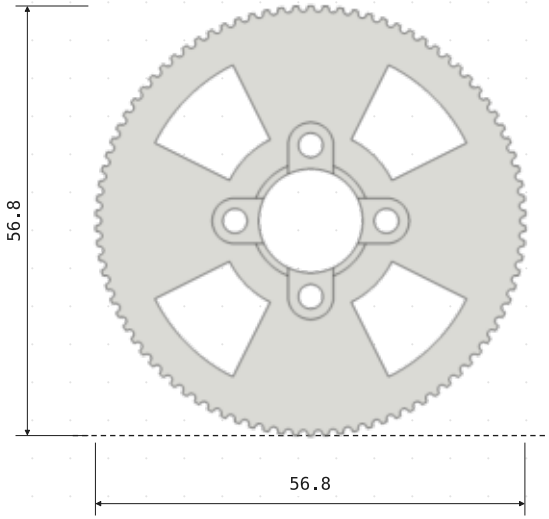
8 Part arm_gear_body



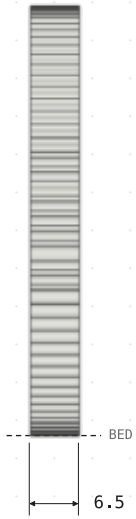
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_gear_body.stl
qty	2
material	PLA
infill	25%
orientation	flat on the bed
size x y z	56.8 x 6.5 x 56.8 mm
volume	11.2 cm3
mass ea	8.7 g (estimate)
scale	1:1 on A4
used in	step –: –

NOTES

[1] Print 2 x in PLA, flat on the bed. The dashed BED line is the print bed.

[2] Overall size 56.8 x 6.5 x 56.8 mm in the print orientation (measured from the STL).

[3] Mass 8.7 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

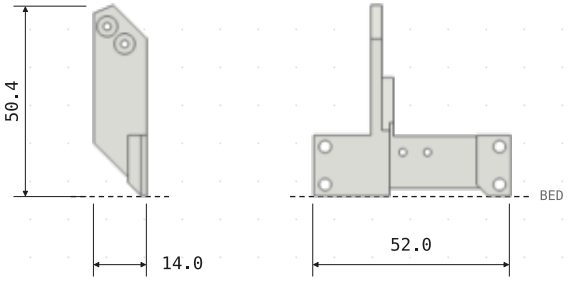
SHEET TITLE
PART ARM_GEAR_BODY

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:1	2026-10-30	8 / 33

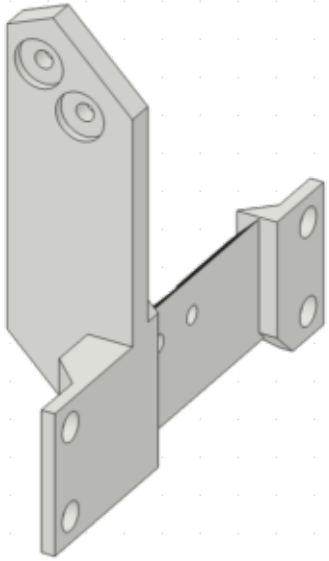
9 Part arm_endstop_bracket



TOP



FRONT RIGHT



ISO (NOT TO SCALE)

PART DATA

FIELD	VALUE
file	stl/arm_endstop_bracket.stl
qty	1
material	PLA
infill	25%
orientation	flat on the bed
size x y z	14.0 x 52.0 x 50.4 mm
volume	3.9 cm3
mass ea	3.7 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat on the bed. The dashed BED line is the print bed.

[2] Overall size 14.0 x 52.0 x 50.4 mm in the print orientation (measured from the STL).

[3] Mass 3.7 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

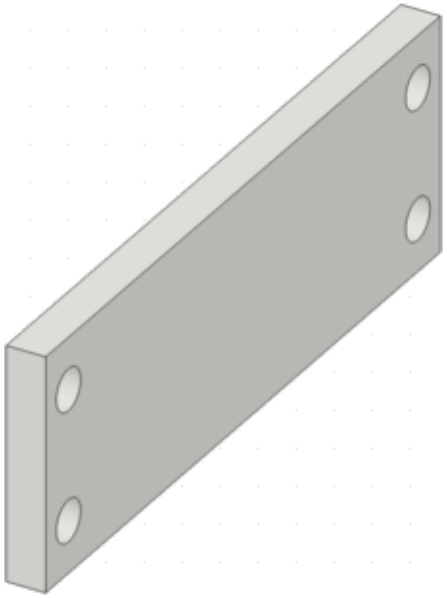
SHEET TITLE
PART ARM_ENDSTOP_BRACKET

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:2	2026-10-30	9 / 33

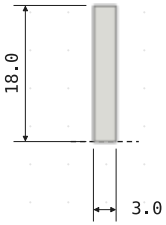
10 Part arm_stabilizer



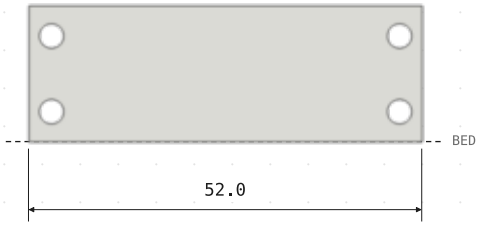
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_stabilizer.stl
qty	1
material	PLA
infill	25%
orientation	flat on the bed
size x y z	3.0 x 52.0 x 18.0 mm
volume	2.7 cm3
mass ea	2.6 g (estimate)
scale	1:1 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat on the bed. The dashed BED line is the print bed.

[2] Overall size 3.0 x 52.0 x 18.0 mm in the print orientation (measured from the STL).

[3] Mass 2.6 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

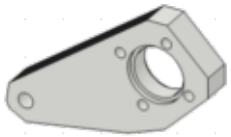
SHEET TITLE
PART ARM_STABILIZER

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:1	2026-10-30	10 / 33

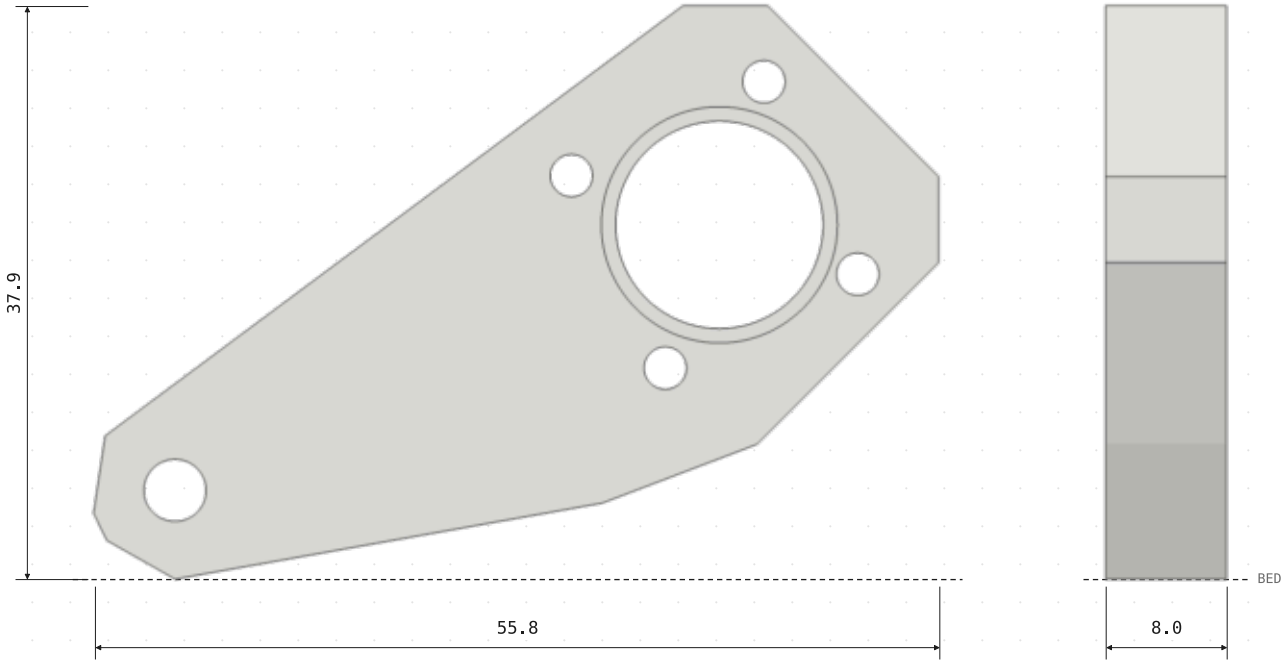
11 Part arm_lever



TOP



ISO (NOT TO SCALE)



FRONT

RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_lever.stl
qty	1
material	PLA
infill	25%
orientation	flat on the bed
size x y z	55.8 x 8.0 x 37.9 mm
volume	7.6 cm3
mass ea	5.2 g (estimate)
scale	2:1 on A4
used in	step –: –

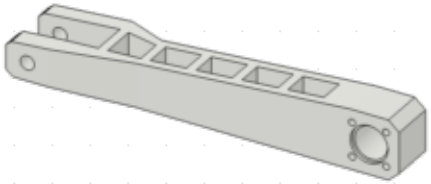
NOTES

[1] Print 1 x in PLA, flat on the bed. The dashed BED line is the print bed.

[2] Overall size 55.8 x 8.0 x 37.9 mm in the print orientation (measured from the STL).

[3] Mass 5.2 g each is computed from volume x density x shell/infill factor (estimate until weighed).

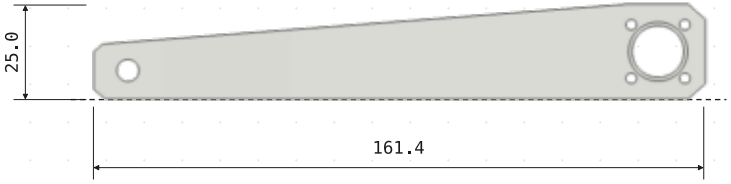
12 Part arm_lower_shank_140



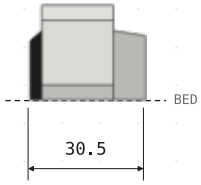
ISO (NOT TO SCALE)



TOP



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_lower_shank_140.stl
qty	1
material	PLA
infill	25%
orientation	flat, 140 mm
size x y z	161.4 x 30.5 x 25.0 mm
volume	40.7 cm3
mass ea	27.3 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat, 140 mm. The dashed BED line is the print bed.

[2] Overall size 161.4 x 30.5 x 25.0 mm in the print orientation (measured from the STL).

[3] Mass 27.3 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

SHEET TITLE
PART ARM_LOWER_SHANK_140

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:2	2026-10-30	12 / 33

13 Part arm_upper_shank_140

PART DATA

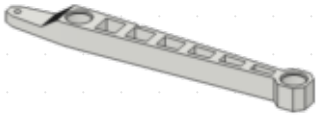
FIELD	VALUE
file	stl/arm_upper_shank_140.stl
qty	1
material	PLA
infill	25%
orientation	flat, 140 mm
size x y z	196.0 x 20.0 x 15.0 mm
volume	26.4 cm3
mass ea	19.6 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat, 140 mm. The dashed BED line is the print bed.

[2] Overall size 196.0 x 20.0 x 15.0 mm in the print orientation (measured from the STL).

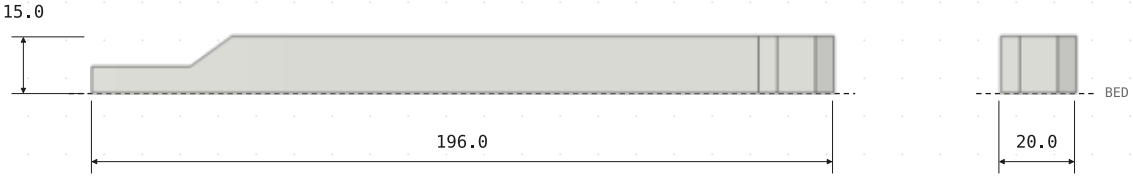
[3] Mass 19.6 g each is computed from volume x density x shell/infill factor (estimate until weighed).





TOP

ISO (NOT TO SCALE)

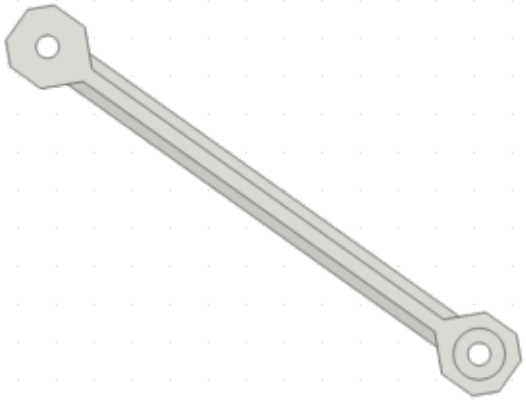


FRONT

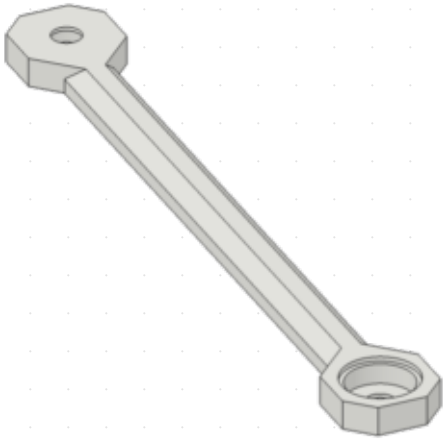
RIGHT

MECHY					SHEET TITLE PART ARM_UPPER_SHANK_140
	PROJECT Mechy Demo	VARIANT BAM–V2	SCALE 1:2	DATE 2026–10–30	SHEET 13 / 33

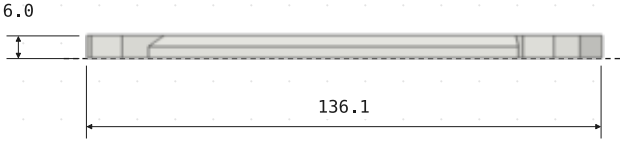
14 Part arm_pleuel_140



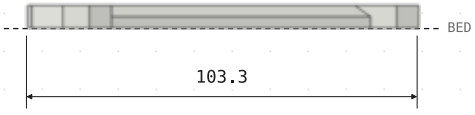
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_pleuel_140.stl
qty	1
material	PLA
infill	25%
orientation	flat
size x y z	136.1 x 103.3 x 6.0 mm
volume	8.7 cm3
mass ea	7.1 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat. The dashed BED line is the print bed.

[2] Overall size 136.1 x 103.3 x 6.0 mm in the print orientation (measured from the STL).

[3] Mass 7.1 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

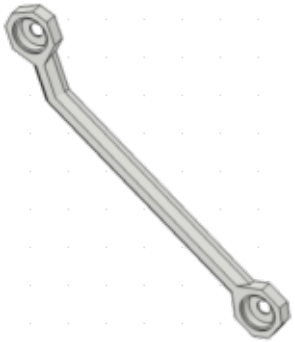
SHEET TITLE
PART ARM_PLEUEL_140

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:2	2026-10-30	14 / 33

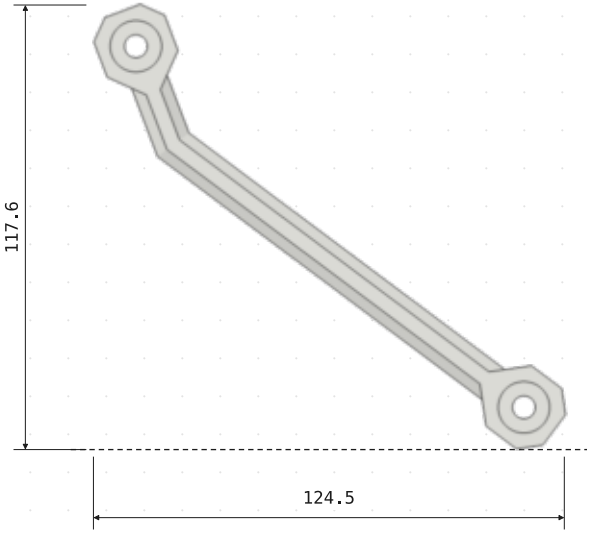
15 Part arm_pleuel_bend_lower_140



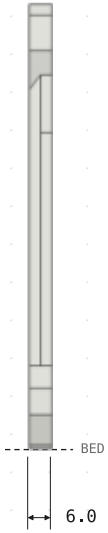
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_pleuel_bend_lower_140.stl
qty	1
material	PLA
infill	25%
orientation	flat
size x y z	124.5 x 6.0 x 117.6 mm
volume	8.9 cm3
mass ea	7.2 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat. The dashed BED line is the print bed.

[2] Overall size 124.5 x 6.0 x 117.6 mm in the print orientation (measured from the STL).

[3] Mass 7.2 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

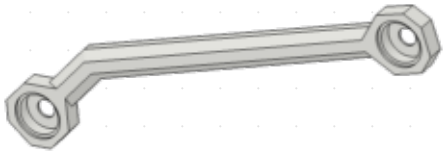
SHEET TITLE
PART ARM_PLEUEL_BEND_LOWER_140

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:2	2026-10-30	15 / 33

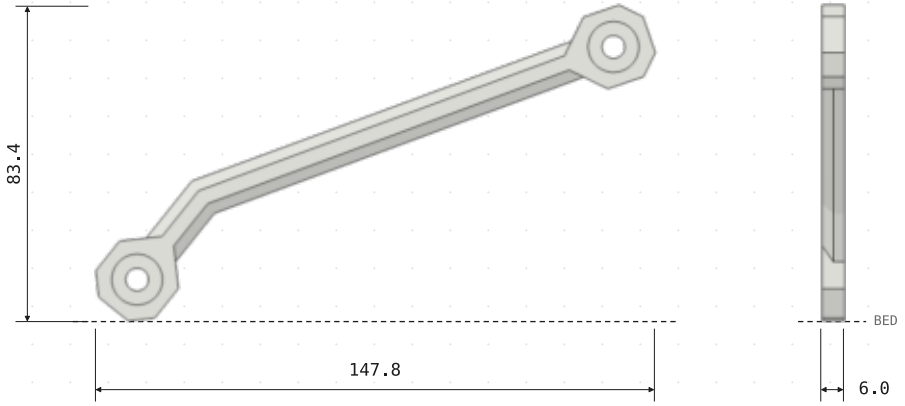
16 Part arm_pleuel_bend_upper_140



TOP



ISO (NOT TO SCALE)



FRONT

RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_pleuel_bend_upper_140.stl
qty	1
material	PLA
infill	25%
orientation	flat; BAM-made part (CC BY 4.0): the vendor 140 mm kit has no upper levelling link
size x y z	147.8 x 6.0 x 83.4 mm
volume	8.9 cm3
mass ea	7.2 g (estimate)
scale	1:2 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat; BAM-made part (CC BY 4.0): the vendor 140 mm kit has no upper levelling link. The dashed BED line is the print bed.

[2] Overall size 147.8 x 6.0 x 83.4 mm in the print orientation (measured from the STL).

[3] Mass 7.2 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

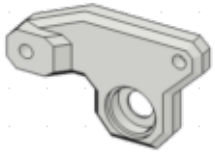
SHEET TITLE
PART ARM_PLEUEL_BEND_UPPER_140

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:2	2026-10-30	16 / 33

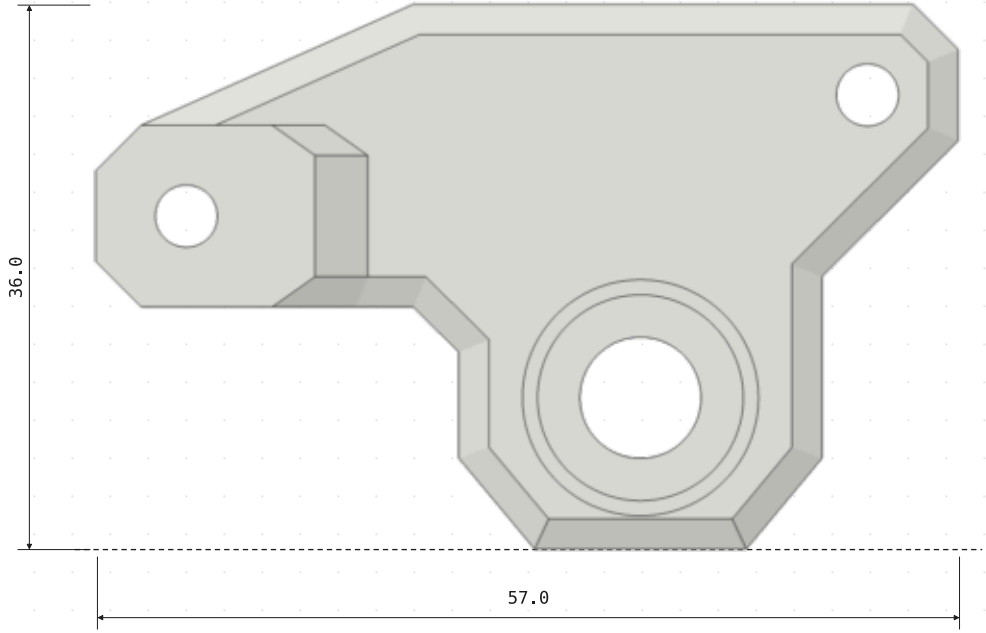
17 Part arm_triplate



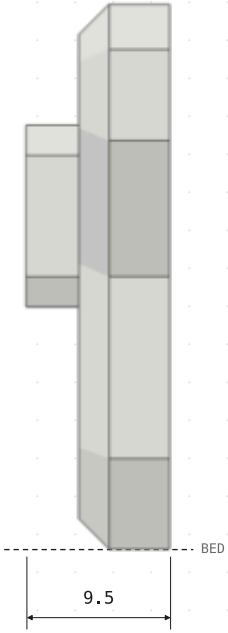
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_triplate.stl
qty	1
material	PLA
infill	25%
orientation	flat
size x y z	57.0 x 9.5 x 36.0 mm
volume	7.1 cm3
mass ea	5.2 g (estimate)
scale	2:1 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat. The dashed BED line is the print bed.

[2] Overall size 57.0 x 9.5 x 36.0 mm in the print orientation (measured from the STL).

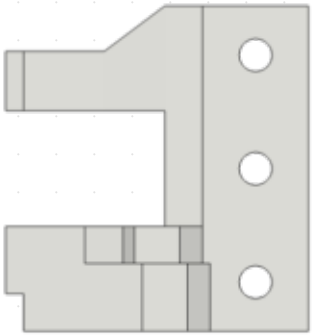
[3] Mass 5.2 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

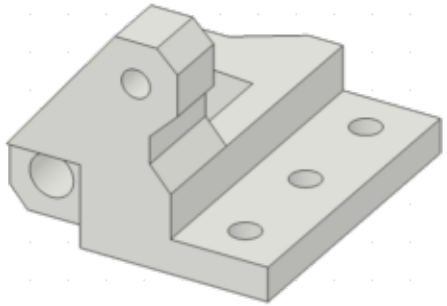
SHEET TITLE
PART ARM_TRIPLATE

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	2:1	2026-10-30	17 / 33

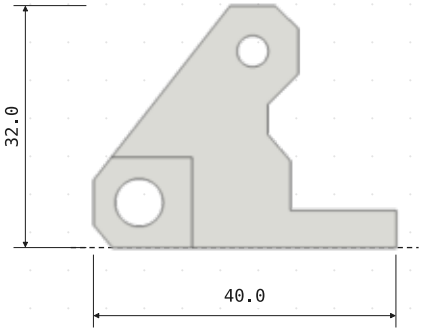
18 Part arm_manipulator



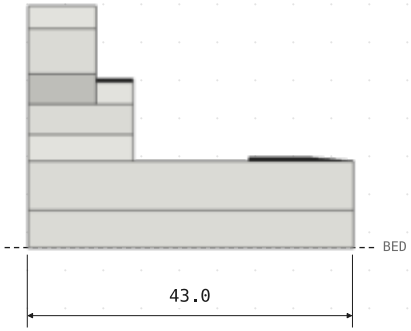
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/arm_manipulator.stl
qty	1
material	PLA
infill	25%
orientation	flat
size x y z	40.0 x 43.0 x 32.0 mm
volume	13.0 cm3
mass ea	8.5 g (estimate)
scale	1:1 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat. The dashed BED line is the print bed.

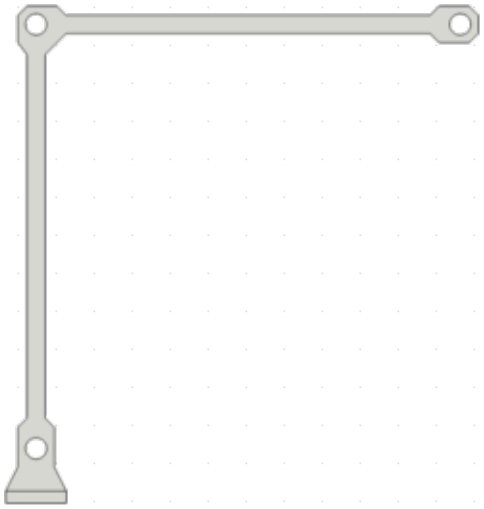
[2] Overall size 40.0 x 43.0 x 32.0 mm in the print orientation (measured from the STL).

[3] Mass 8.5 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

SHEET TITLE
PART ARM_MANIPULATOR

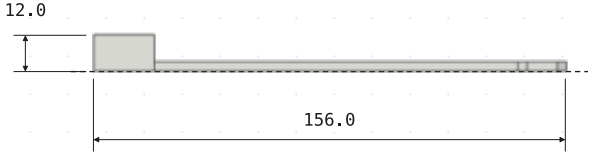
PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:1	2026-10-30	18 / 33



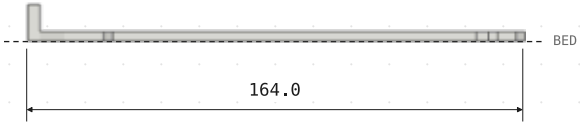
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/tool_shank_calibrator_140.stl
qty	1
material	PLA
infill	25%
orientation	flat; homing aid
size x y z	156.0 x 164.0 x 12.0 mm
volume	7.0 cm3
mass ea	7.3 g (estimate)
scale	1:2.5 on A4
used in	step –: –

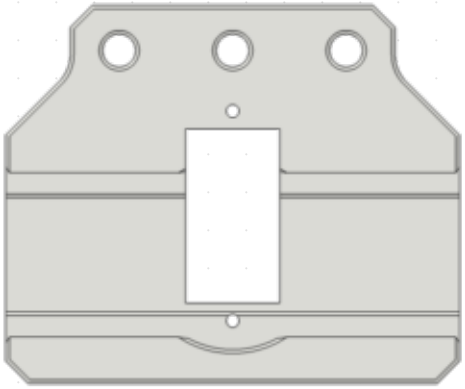
NOTES

[1] Print 1 x in PLA, flat; homing aid. The dashed BED line is the print bed.

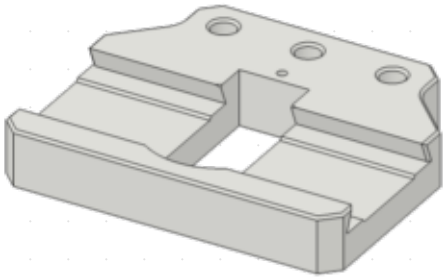
[2] Overall size 156.0 x 164.0 x 12.0 mm in the print orientation (measured from the STL).

[3] Mass 7.3 g each is computed from volume x density x shell/infill factor (estimate until weighed).

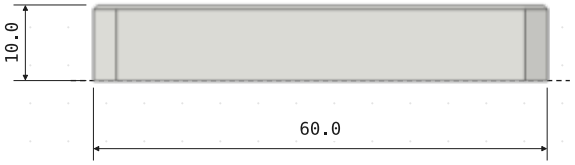
20 Part grip_base



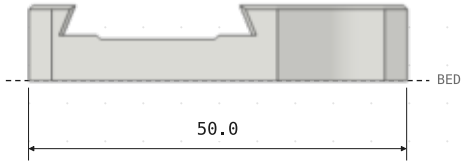
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/grip_base.stl
qty	1
material	PLA
infill	25%
orientation	for the MG90S / 9 g servo
size x y z	60.0 x 50.0 x 10.0 mm
volume	19.2 cm3
mass ea	12.1 g (estimate)
scale	1:1 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, for the MG90S / 9 g servo. The dashed BED line is the print bed.

[2] Overall size 60.0 x 50.0 x 10.0 mm in the print orientation (measured from the STL).

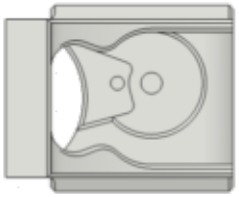
[3] Mass 12.1 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

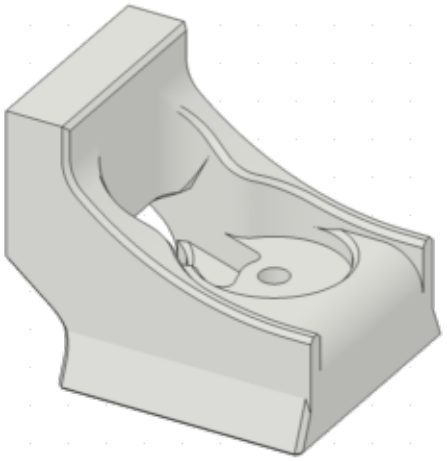
SHEET TITLE
PART GRIP_BASE

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:1	2026-10-30	20 / 33

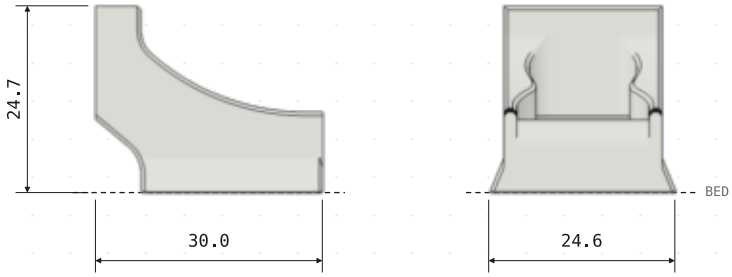
21 Part grip_finger



TOP



ISO (NOT TO SCALE)



FRONT

RIGHT

PART DATA

FIELD	VALUE
file	stl/grip_finger.stl
qty	2
material	PLA
infill	25%
orientation	flat
size x y z	30.0 x 24.6 x 24.7 mm
volume	6.4 cm3
mass ea	4.5 g (estimate)
scale	1:1 on A4
used in	step –: –

NOTES

[1] Print 2 x in PLA, flat. The dashed BED line is the print bed.

[2] Overall size 30.0 x 24.6 x 24.7 mm in the print orientation (measured from the STL).

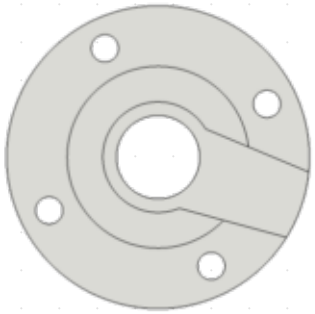
[3] Mass 4.5 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

SHEET TITLE
PART GRIP_FINGER

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	1:1	2026-10-30	21 / 33

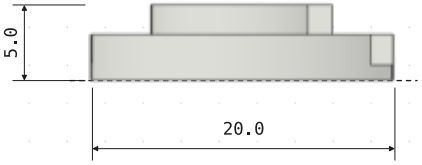
22 Part grip_shaft



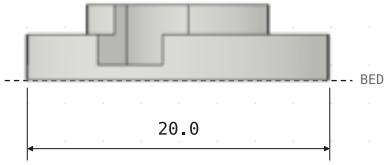
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/grip_shaft.stl
qty	1
material	PLA
infill	25%
orientation	flat
size x y z	20.0 x 20.0 x 5.0 mm
volume	0.9 cm3
mass ea	1.0 g (estimate)
scale	2:1 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat. The dashed BED line is the print bed.

[2] Overall size 20.0 x 20.0 x 5.0 mm in the print orientation (measured from the STL).

[3] Mass 1.0 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

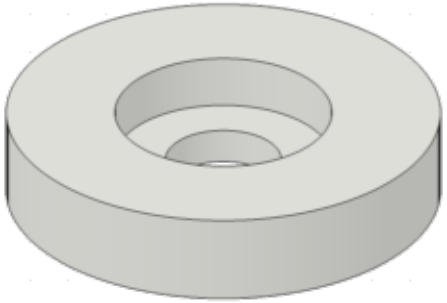
SHEET TITLE
PART GRIP_SHAFT

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	2:1	2026-10-30	22 / 33

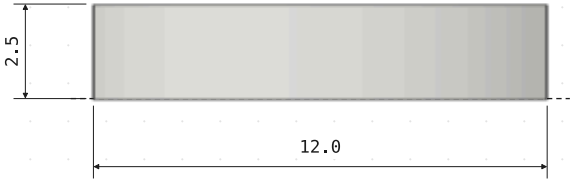
23 Part grip_screw_secure



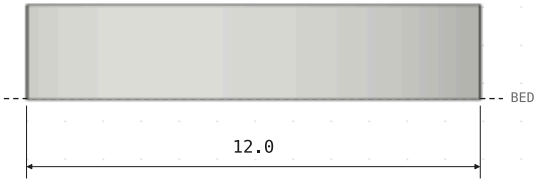
TOP



ISO (NOT TO SCALE)



FRONT



RIGHT

PART DATA

FIELD	VALUE
file	stl/grip_screw_secure.stl
qty	1
material	PLA
infill	25%
orientation	flat
size x y z	12.0 x 12.0 x 2.5 mm
volume	0.2 cm3
mass ea	0.3 g (estimate)
scale	5:1 on A4
used in	step –: –

NOTES

[1] Print 1 x in PLA, flat. The dashed BED line is the print bed.

[2] Overall size 12.0 x 12.0 x 2.5 mm in the print orientation (measured from the STL).

[3] Mass 0.3 g each is computed from volume x density x shell/infill factor (estimate until weighed).

MECHY

SHEET TITLE
PART GRIP_SCREW_SECURE

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	5:1	2026-10-30	23 / 33

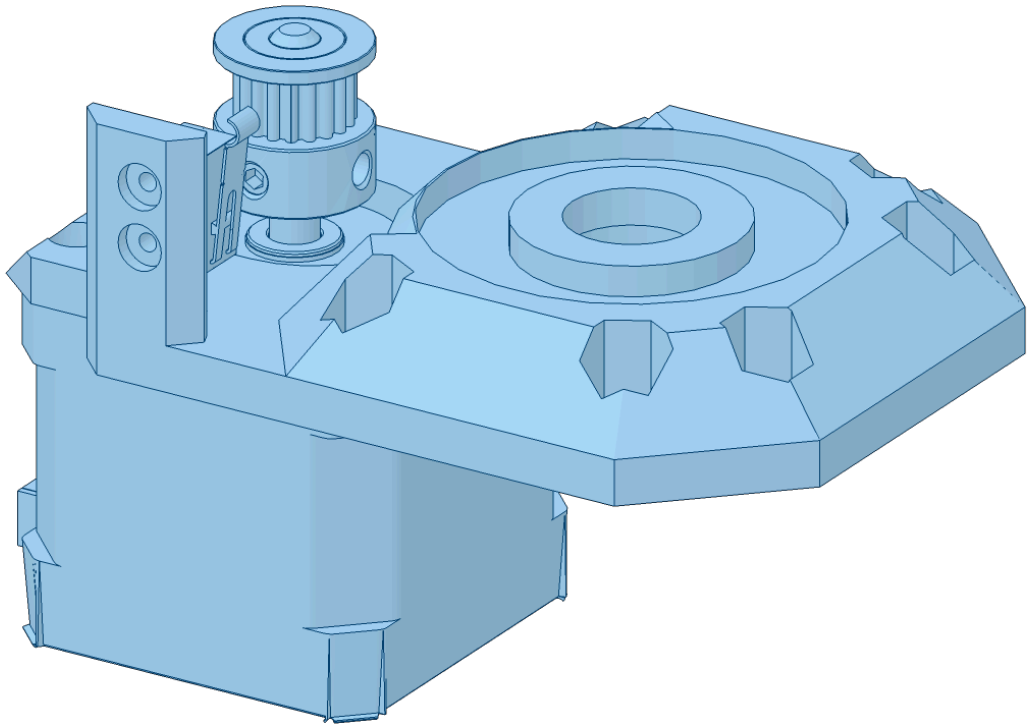
24 Step 1 Prepare the printed parts

DO

1. Print every part in the print list (21 pieces, about 206 g PLA) in the listed orientation.
2. Clean every bearing bore (F624ZZ / F686ZZ press fits) with a drill bit until a bearing presses in by hand; do not hammer printed seats.
3. Print the 140 mm shank calibrator too - it sets the homing steps in the last step.

MECHY	SHEET TITLE: STEP 1 PREPARE THE PRINTED PARTS				
	PROJECT: Mechy Demo	VARIANT: BAM-V2	SCALE: NTS	DATE: 2026-10-30	SHEET: 24 / 33

25 Step 2 Socket and base yaw drive



BLUE = ADDED IN THIS STEP · GREY = ALREADY ASSEMBLED · ARM IN THE READY POSE

PRINTED PARTS

PART	QTY
robot_belt_arm socket.stl	1

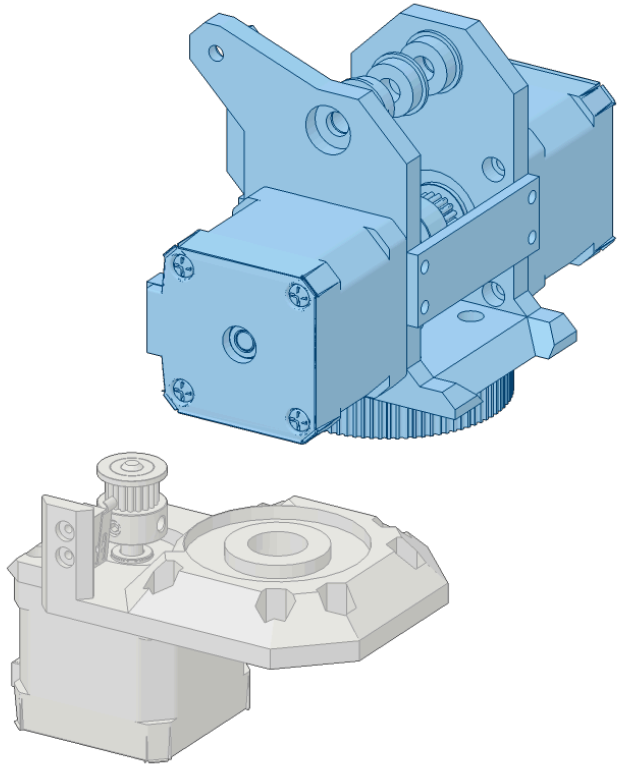
HARDWARE (SCREWS, BEARINGS, BOUGHT PARTS)

ITEM	QTY
NEMA 17 (yaw)	1
GT2 20T pulley	1
yaw endstop	1
51105 thrust bearing	1
Endstop microswitch	3
M2x10 screw + nut	6

DO

1. Seat the 51105 thrust bearing in the socket and press the socket into its stand.
2. Mount the yaw NEMA 17 inside the socket with its 20T pulley (set screw on the flat).
3. Fit the Z endstop microswitch and wire it to the shield's Z limit header (D11).

26 Step 3 Main body, motors and belts



BLUE = ADDED IN THIS STEP (MOVED OUT FOR CLARITY) · GREY = ALREADY ASSEMBLED · ARM IN THE READY POSE

PRINTED PARTS

PART	QTY
robot_belt_arm rotategear.stl	1
robot_belt_arm main_body.stl	1
robot_belt_arm stabilizer.stl	1

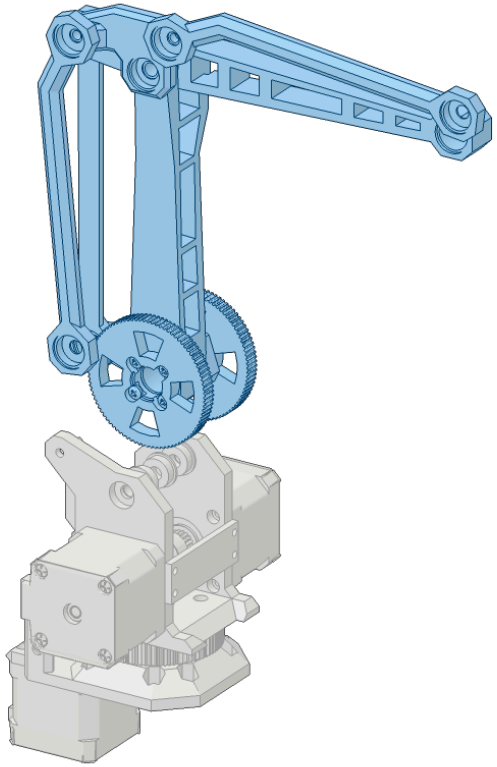
HARDWARE (SCREWS, BEARINGS, BOUGHT PARTS)

ITEM	QTY
NEMA 17	2
GT2 20T pulley	2
endstop bracket	1
bearing F624ZZ	4
GT2 pulley 20T	3
GT2 closed belt 200 mm	3

DO

1. Loop the base GT2 belt over the rotategear and lower the main body onto the 51105.
2. Mount the two arm NEMA 17s on the main body (Y = lower shank, X = upper/lever), 20T pulleys on both, belts looped but not tensioned.
3. Bolt on the endstop bracket and stabilizer; fit the Y and X endstop switches.

27 Step 4 Linkage: shanks, lever, pleuel and levelling links



BLUE = ADDED IN THIS STEP (MOVED OUT FOR CLARITY) · GREY = ALREADY ASSEMBLED · ARM IN THE READY POSE

PRINTED PARTS

PART	QTY
robot_belt_arm gear_body.stl	2
robot_belt_arm pleuel.stl	1
robot_belt_arm lower_shank.stl	1
robot_belt_arm pleuel_bend_upper.stl	1
robot_belt_arm pleuel_bend_lower.stl	1
robot_belt_arm triplate.stl	1
robot_belt_arm lever.stl	1
robot_arm upper_shank.stl	1

HARDWARE (SCREWS, BEARINGS, BOUGHT PARTS)

ITEM	QTY
screw	8
bearing F686ZZ	2
F624ZZ flanged bearing	6
F686ZZ flanged bearing	12
M3x6 screw	36
M3 nut	8
M4x10 screw	6
M4x16 screw	8
M4x20 screw	6
M4 nut	14
M6x20 screw	1
M6x40 screw	2
M6x50 screw	1
M6 nut	4

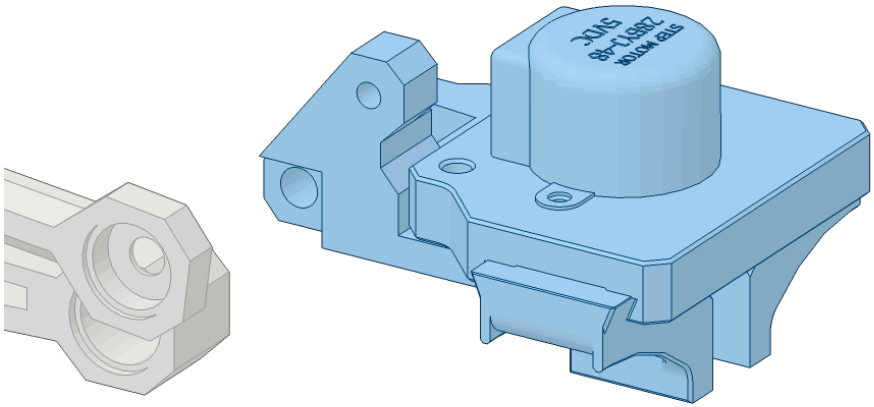
DO

1. Shoulder S: lower shank + lower 90T gear + lever + upper 90T gear share the S axis (F624ZZ bearings). Free rotation, no play.
2. Pleuel: lever end L to the upper shank's drive pin (F624ZZ); it must stay parallel to the lower shank.
3. Elbow E: lower shank + upper shank + triplate on one axis (F686ZZ).
4. Lower levelling link: B0 on the main body to the triplate's rear pin; upper levelling link (BAM part) triplate T2 to the manipulator T3.
5. Wrist W: manipulator on the upper shank end (F686ZZ). The linkage keeps it level.

MECHY

SHEET TITLE: STEP 4 LINKAGE: SHANKS, LEVER, PLEUEL AND LEVELLING LINKS

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM-V2	NTS	2026-10-30	27 / 33



BLUE = ADDED IN THIS STEP (MOVED OUT FOR CLARITY) · GREY = ALREADY ASSEMBLED · ARM IN THE READY POSE

PRINTED PARTS

PART	QTY
toolheads_stl/gripper_stl_by_ftobler/gripperFinger_by_ftobler.stl	2
gripper	3
toolheads_stl/gripper_stl_by_ftobler/gripper_shaft.stl	1
toolheads_stl/gripper_stl_by_ftobler/gripperBase_by_ftobler.stl	1
robot_belt_arm manipulator.stl	1

HARDWARE (SCREWS, BEARINGS, BOUGHT PARTS)

ITEM	QTY
gripper motor (28BYJ; ref_arm uses a servo)	1
Servo MG90S (gripper toolhead)	1

DO

1. Cup: bolt the cup flange to the manipulator plate (M3 into the printed flange; the cup opening points straight up).
2. Stick the foam liner ring inside the cup wall.

29 Step 6 Electronics

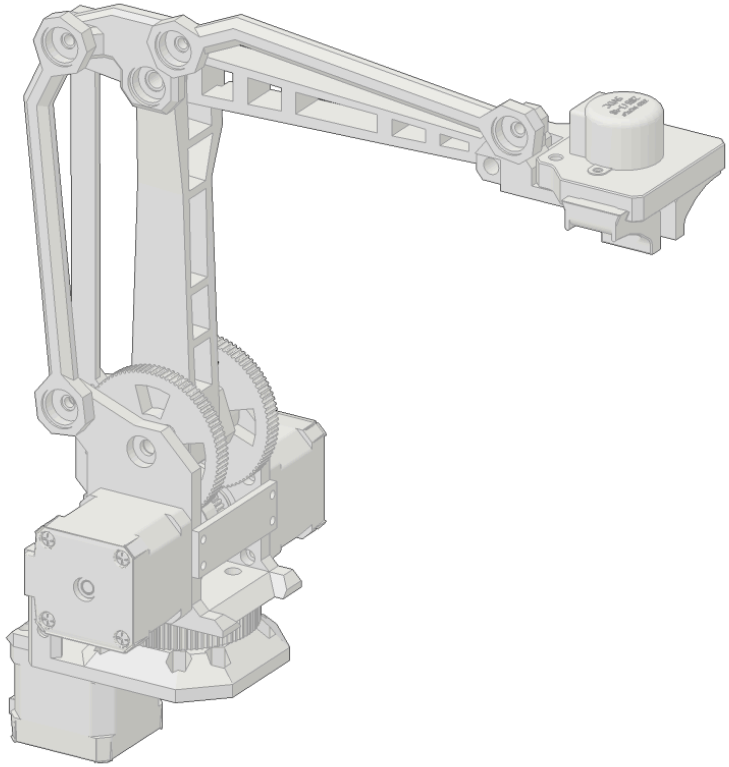
HARDWARE (SCREWS, BEARINGS, BOUGHT PARTS)

ITEM	QTY
Arduino Uno R3	1
CNC Shield V3	1
A4988 stepper driver	3
A4988 stepper driver (spare)	1
Power supply 24 V 6 A	1
Electrolytic capacitor 100 uF 50 V	1
Stepper cable	3
Silicone wire 18 AWG	2

DO

1. Plug the three A4988 drivers into X/Y/Z (EN to EN), set MS1/MS2/MS3 jumpers for the chosen microsteps and Vref BEFORE connecting the belts (wiring sheet).
2. Motors: X = upper arm, Y = lower arm, Z = yaw. Endstops to the matching limit headers.
3. 100 uF capacitor across the shield's motor power input, 24 V supply, Arduino on USB. Never unplug a motor with the supply on.
4. Flash the generated firmware (flashing steps on the wiring sheet).

30 Step 7 Home and calibrate



BLUE = BELTS TO TENSION · ARM IN THE READY POSE

DO

1. Mount the shank calibrator and set X_HOME_STEPS / Y_HOME_STEPS / Z_HOME_STEPS per the control README (the generated values are estimates until calibrated).
2. Home order: shoulder, elbow, base (the wrist is passive - the linkage levels it).
3. Send HOME with a hand on the power switch; each joint must stop cleanly at its endstop.

Belts, tensioning and homing

BELTS (PITCH LENGTHS, COMPUTED)

STAGE	PULLEYS	CENTRE	COMPUTED	STOCK	TENSION
base	20T–90T	0.0	200.0	200	vendor GT2 200 mm closed loop
shoulder	20T–90T	0.0	200.0	200	vendor
elbow	20T–90T	0.0	200.0	200	vendor

TENSIONING

[1] base: slide the motor in its slots until the 200.0 mm belt deflects about 2.0 mm under a light finger press mid-span (about 1/64 of the span, a common GT2 rule of thumb; estimate), then tighten.

[2] shoulder: slide the motor in its slots until the 200.0 mm belt deflects about 2.0 mm under a light finger press mid-span (about 1/64 of the span, a common GT2 rule of thumb; estimate), then tighten.

[3] elbow: slide the motor in its slots until the 200.0 mm belt deflects about 2.0 mm under a light finger press mid-span (about 1/64 of the span, a common GT2 rule of thumb; estimate), then tighten.

[4] Link belts (520 mm): stock loops sized exactly to the 220 mm centre distance; take up slack with the slotted shaft holes (+-3 mm) before clamping the collars.

[5] Home in this order: shoulder, elbow, base, wrist. Send HOME with a hand on the power switch and check each joint stops at its sensor (table on the belts and homing sheet).

HOMING SENSORS (ONE PER JOINT)

JOINT	SENSOR	WHERE / TRIGGER	HOMES
base	endstop microswitch (vendor, on the socket)	socket; rotategear tab	q0 at the Z endstop; calibrate with Z_HOME_STEPS
shoulder	endstop microswitch (vendor)	main body; lower shank	a1 = q1 at the Y endstop; recalibrate with tool_shank_calibrator_140
elbow	endstop microswitch (vendor)	main body; lever	a2 = q1+q2 at the X endstop; recalibrate with the calibrator
wrist	none (passive joint: the linkage holds q3 = −(q1+q2))	–; –	not homed (no motor)

HOMING ORDER

shoulder > elbow > base > wrist

JOINT TO MOTOR MAPPING

[1] belt pass-through: the elbow motor sets the forearm's absolute pitch q1+q2 and the wrist motor the hand's q1+q2+q3; moving the shoulder alone keeps forearm and cup pitch fixed

[2] motor_angle = gear_ratio * (C @ q); steps = round(steps_per_rad * (C @ q)); q = C^-1 @ (motor_angle / gear_ratio); motor torque = (C^-T tau_joint) / gear_ratio

[3] Steps per degree of the motor-side angle at 8 microsteps: base 25, shoulder 25, elbow 25, wrist 20

MECHY

SHEET TITLE: BELTS, TENSIONING AND HOMING

PROJECT	VARIANT	SCALE	DATE	SHEET
Mechy Demo	BAM–V2	NTS	2026–10–30	31 / 33

32 Wiring and flashing

CONTROL NOT GENERATED

[1] No control/bam-v2/README.md yet. Run `mechy control gen bam-v2` to generate the firmware, host code and wiring tables, then rebuild this guide.

33 Checks, not verified, open decisions

CHECKS RUN WHILE BUILDING THIS GUIDE

CHECK	EXPECTED	GOT	STATUS
BOM total on this guide = bom.csv	$131.30	$131.30	PASS
Every print-list part has an STL and a part sheet	19	19 sheets, 0 missing STLs	PASS
Printed pieces: print list = CAD	21	21	PASS
Every CAD part is installed in exactly one step	44	31	FAIL
Every BOM line is used by a step	28 lines	missing: NEMA 17 stepper 42SHDC3026-15B, NEMA 17 stepper 42SHDC3026-15B (spare)	FAIL
CAD outputs match params.yaml	b03a5e546b30	mass_props.json b03a5e546b30	PASS

NOT VERIFIED

[1] belt tensioning and tooth skipping, printed-surface quality, joint stiffness/backlash, infill-based masses (estimate — weigh the parts), home sensor positions (estimates, calibrate). Open decisions: the 4th motor/driver — vendor rail module or a wrist servo (spare). (design/bam-v2/summary.md)

OPEN DECISIONS

[1] Every BOM line is used by a step: expected 28 lines, missing: NEMA 17 stepper 42SHDC3026-15B, NEMA 17 stepper 42SHDC3026-15B (spare). Fix before building. (check)

[2] Every CAD part is installed in exactly one step: expected 44, 31. Fix before building. (check)

[3] the 4th motor/driver — vendor rail module or a wrist servo (spare). (design/bam-v2/summary.md)

[4] Choose a concept to develop for the 600 g arm. A is the conventional first option; A and B are the proposed CAD/simulation comparison pair, with C available if you prefer counterweights. No new CAD until your pick. All retain four owned motors, wrist roll, MG90S, black finish, X1E fit target and PETG exceptions. Grip, exact motor/step-rate capability and delivered cost below $100 remain unverified; choosing a concept does not approve fabrication or a budget increase. (options/arm-600g-concept.yaml)

[5] No firmware yet: run `mechy control gen bam-v2`. (control)